CP317 - Software Engineering

Project Title: Car Rental System

Group ID: Team # 28

Team Members and Roles:

- SabilUllah Hussaini (Product Owner)
- Moosa Ahmed (Scrum Master)
- Mahad Farrukh (Developer)
- Ayesha Raheel (Developer)
- Kritika Kamath (Developer)
- Sahil Sandhu (Developer)
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Abstract

Our car rental service aims to satisfy the convenience and flexibility demands of today's tourists. Our web-based application will make it easy for customers to search, reserve, and manage their rental cars. It will provide the rental company with strong capabilities to monitor business performance, manage fleet inventory, and comprehend consumer preferences. In order to meet the obvious market need for digital solutions in the transportation industry, our team will create a safe, easy-to-use, and effective system that expedites the whole vehicle rental process from search to return.

Project Description & Objectives

Description:

Our team will design and implement a comprehensive Car Rental System. Customers will be able to make and manage reservations, check their rental history, and search for available cars based on a variety of parameters (such as kind, date, and price) via this system. At the same time, it will give employees of the rental company an administrative interface to view past and current rentals, manage the fleet of vehicles, and create reports on business indicators.

Objectives:

1. Give clients an easy-to-use interface so they can look for available vehicles and book online.
2. Give users the option to register so they can view a history of past rents, modify their personal information, and check their current reservations.
3. Give administrators of rental companies the ability to add, edit, and remove cars from the fleet.
4. Give administrators access to a dashboard where they may see daily rental summaries, upcoming returns, and all existing rentals.
5. Provide management with analytical reports that highlight common car models and rental patterns to help guide inventory and business choices.
6. Use secure data handling procedures and authentication to guarantee the safety of user and payment information.

Initial Product Backlog (Draft)

Story ID	Feature Category	User Story	Actor
AUTH-1	Authentication	Enables a new customer to register for an account using an email address and password, granting them the ability to make and monitor vehicle reservations.	Custom
AUTH-2	Authentication	Allows a registered customer to securely log in to the system to access their personalized account dashboard.	Custom
UI-1	Search & Booking	Provides customers with the functionality to filter the available vehicle fleet based on rental dates, preferred car category, and budget.	Custom
UI-2	Search & Booking	Allows a customer to finalize their selection by reserving a specific vehicle for a chosen date range, ensuring the car is secured for their use.	Custom
UI-3	User Dashboard	Gives customers a centralized view of their reservation history and upcoming bookings for easy planning and management.	Custom
ADMIN-1	Fleet Management	Empowers administrators to add new vehicles to the system and modify existing vehicle details (e.g., status, mileage) to maintain an accurate and up-to-date inventory.	Admins
ADMIN-2	Operations	Provides administrators with an overview dashboard of all current and future rentals to efficiently track and manage business operations.	Admins

Ethical Considerations

1. Data Privacy and Security:

- ***Issue:*** Customers' names, addresses, driver's license numbers, and payment details are among the private data that the system will collect and retain. Identity theft and financial fraud could be the outcome of a data leak.
- ***Mitigation:*** All private information will be encrypted in the database while it is at rest and while it is in transit (using HTTPS/TLS). A powerful algorithm (such as bcrypt) will be used to hash and salt the passwords. According to the least privilege principle, only authorized administrative staff will have access to PII. We'll assess security on a frequent basis.

2. Accessibility:

- ***Issue:*** If the web application is not designed with accessibility in mind, it could prevent individuals with disabilities (e.g., visual, motor) from using the service, constituting discrimination and excluding a significant portion of the market.
- ***Mitigation:*** The concepts of WCAG (Web Content Accessibility Guidelines) will guide our design and development. Screen reader compatibility (using semantic HTML and ARIA labels), keyboard navigability, appropriate contrast ratios, and text replacements for non-text material are all part of this. To confirm compliance, we'll employ both automatic and manual testing methods.

3. Algorithmic Fairness and Bias:

- ***Issue:*** The algorithms used for dynamic pricing or for determining rental eligibility (e.g., based on age or driving record) could inadvertently introduce bias. For example, pricing might disproportionately affect users from certain regions, or eligibility rules might unfairly exclude specific demographic groups.
- ***Mitigation:*** We will consciously design pricing algorithms to be based on clear, objective factors like demand, vehicle type, and rental duration, avoiding proxies for demographic data. Eligibility criteria will be reviewed to ensure they are directly related to the legitimate business goal of risk assessment (e.g., valid driver's license, insurance) and are applied consistently to all users. We will document the logic behind these systems for transparency.

Team Blog (Excel Template)

Note: This would be a separate Excel file. The entry for this milestone would include the team information from the cover page and a confirmation that the blog will be maintained.

File to be submitted separately: Team-28-Blog.xlsx